

DATASCI 350 — Data Science Computing

Final Project Report

Population Dynamics in Asia:
Trends in Life Expectancy, Under-5 Mortality,
and Adolescent Fertility (1990–2024)

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1. Introduction

Over the past three decades, the Asia-Pacific region has undergone one of the most rapid demographic transformations in modern history. Rising incomes, expanding healthcare systems, improving female education, and targeted public health programmes have together reshaped the life prospects of billions of people across the region — from the high-income economies of East Asia to the rapidly developing nations of South Asia and the Pacific.

This project analyses **population dynamics** across 36 Asian and Pacific countries from **1990 to 2024**, using three key indicators from the World Bank's World Development Indicators (WDI) database: (1) *life expectancy at birth*, (2) *under-5 mortality rate*, and (3) *adolescent fertility rate*. Together these indicators capture how long people live, how many young children survive, and reproductive behaviour among the youngest women of childbearing age — all central dimensions of demographic change.

Research questions:

1. How have life expectancy, under-5 mortality, and adolescent fertility evolved across Asian countries since 1990?
2. Are there diverging or converging trajectories between high-income and lower-income Asian economies?
3. What is the relationship between adolescent fertility and life expectancy, and between child mortality and life expectancy, across countries?

Main findings. All three indicators improved substantially across the region. Life expectancy rose on average by roughly 10 years, under-5 mortality fell dramatically — particularly in South Asia — and adolescent fertility declined across most countries. However, convergence is incomplete: a persistent gap separates Japan and South Korea from lower-income Pacific and South Asian nations. Both adolescent fertility and child mortality show strong negative correlations with life expectancy, consistent with the demographic transition framework.

2. Data Description

2.1 Source and Coverage

Data were obtained directly from the World Bank WDI API using Python's requests library. Countries were drawn from two regional classifications — East Asia & Pacific (EAS) and South Asia (SAS) — yielding **36 countries**. The three indicators retrieved are summarised below.

Indicator	WDI Code	Unit
Life Expectancy at Birth	SP.DYN.LE00.IN	Years
Under-5 Mortality Rate	SH.DYN.MORT	Per 1,000 live births
Adolescent Fertility Rate	SP.ADO.TFRT	Per 1,000 women aged 15–19

Table 1. World Bank WDI indicators used in this analysis.

The final dataset spans **1990–2024** (35 years), with 1,260 observations (36 countries × 35 years). Time series visualisations focus on six representative countries — Japan, South Korea, China, India, Indonesia, and Bangladesh — chosen to span the full spectrum of the region's income and development levels.

2.2 Data Merging

The three indicator datasets were fetched from the WDI API separately and merged on two keys: `country_code` (ISO 3166-1 alpha-3) and `year`. An inner join was used so that only country-year observations with all three indicators present are retained. Country names were mapped from a reference dataframe built from the API's regional country lists.

2.3 Data Cleaning and Pre-processing

Raw API responses included aggregate regional entries (e.g., 'East Asia & Pacific'). These were filtered out by retaining only records with ISO codes of exactly three characters. Missing values within each country's time series were filled by within-group linear interpolation (`df.groupby('country_code').transform(lambda x: x.interpolate())`). Rows still containing NaN after interpolation were dropped. The cleaned dataset was exported as `asian_population_clean.csv`.

2.4 Summary Statistics (via SQL)

Descriptive statistics were computed by loading the cleaned CSV into an in-memory SQLite database and running structured queries. Table 2 reports average values and the range of improvement for selected countries.

Country	Avg. Life Exp. (yrs)	Life Gain (yrs)	Avg. Mort. U5	Mort. Reduction	Avg. Teen Fertility
Japan	82.12	5.72	3.88	3.9	4.19
Korea, Rep.	78.54	12.03	6.34	12.9	2.22
China	74.23	10.05	25.19	47.9	9.42
Malaysia	74.11	5.87	9.87	8.6	12.69
Thailand	73.49	8.68	18.27	27.9	44.31
Indonesia	67.57	8.04	41.97	65.5	47.18
Philippines	68.09	5.71	34.38	29.3	47.20
India	65.59	13.62	70.85	100.4	69.56
Bangladesh	65.69	21.15	68.62	116.0	121.98
Cambodia	64.06	15.61	65.68	101.1	56.23

Table 2. Country-level summary statistics (1990–2024). Life Gain = max – min life expectancy. Mort. Reduction = max – min under-5 mortality rate. Source: SQL queries on the cleaned WDI dataset.

3. Data Analysis

3.1 Life Expectancy Trends (1990–2024)

Figure 1 traces life expectancy at birth for Japan, South Korea, China, India, Indonesia, and Bangladesh from 1990 to 2024. All six countries show continuous improvement. Japan, starting at 78.8 years, remained the regional leader, reaching 84.0 by 2023. South Korea's rise is the most striking — from 71.6 in 1990 to 83.4 in 2023 — nearly closing the gap with Japan in just 33 years. China and Indonesia recorded steady mid-level gains, while India and Bangladesh, both starting below 60 years, delivered the steepest absolute improvements. Bangladesh rose by nearly 19 years over the period, reflecting major advances in child health, maternal care, and female education. A visible dip around 2020–2021 is consistent with excess mortality from the COVID-19 pandemic.

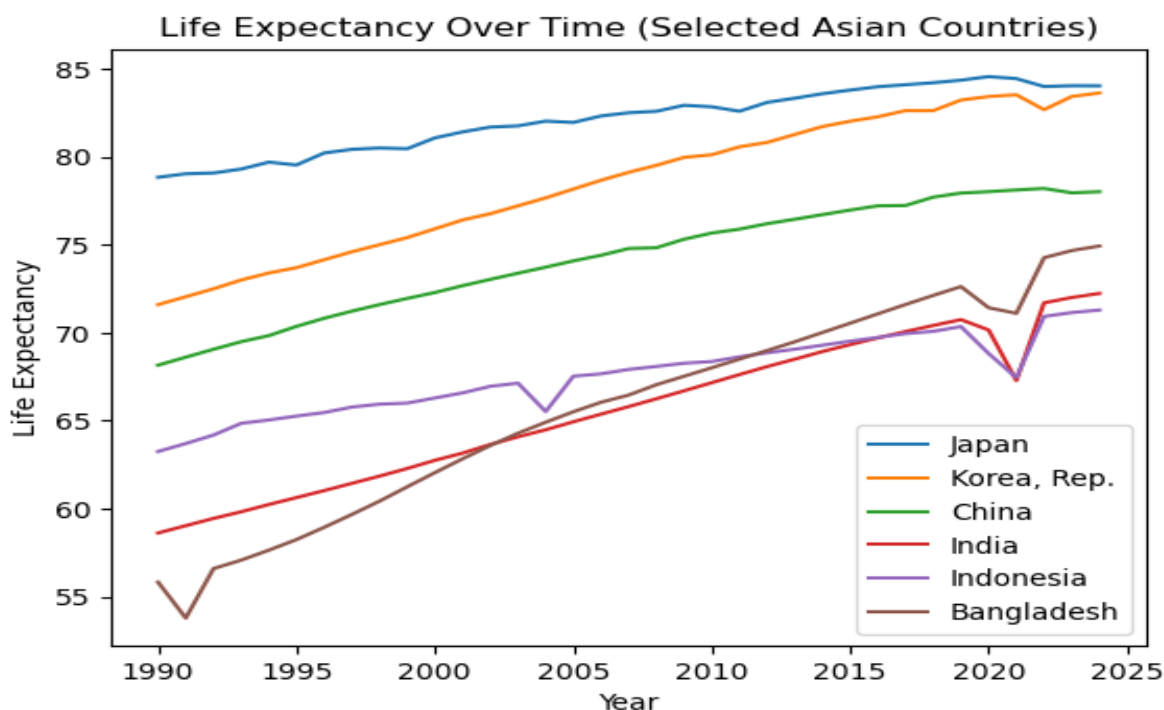


Figure 1. Life expectancy at birth (years), 1990–2024, for six selected Asian countries. Source: World Bank WDI (SP.DYN.LE00.IN).

3.2 Adolescent Fertility Rate Trends

Figure 2 shows the adolescent fertility rate (births per 1,000 women aged 15–19). Japan and South Korea maintained exceptionally low rates throughout — consistently below 5 — reflecting highly developed educational systems and delayed marriage norms. Bangladesh had the highest rate in 1990 at 178 births per 1,000 women, but achieved one of the largest declines, falling to approximately 73 by 2023. India and Indonesia also declined substantially. China's adolescent fertility was low throughout. Thailand is a notable exception: despite its middle-income status, its adolescent fertility remained elevated (around 25–50), suggesting that economic development alone does not fully determine

reproductive behaviour.

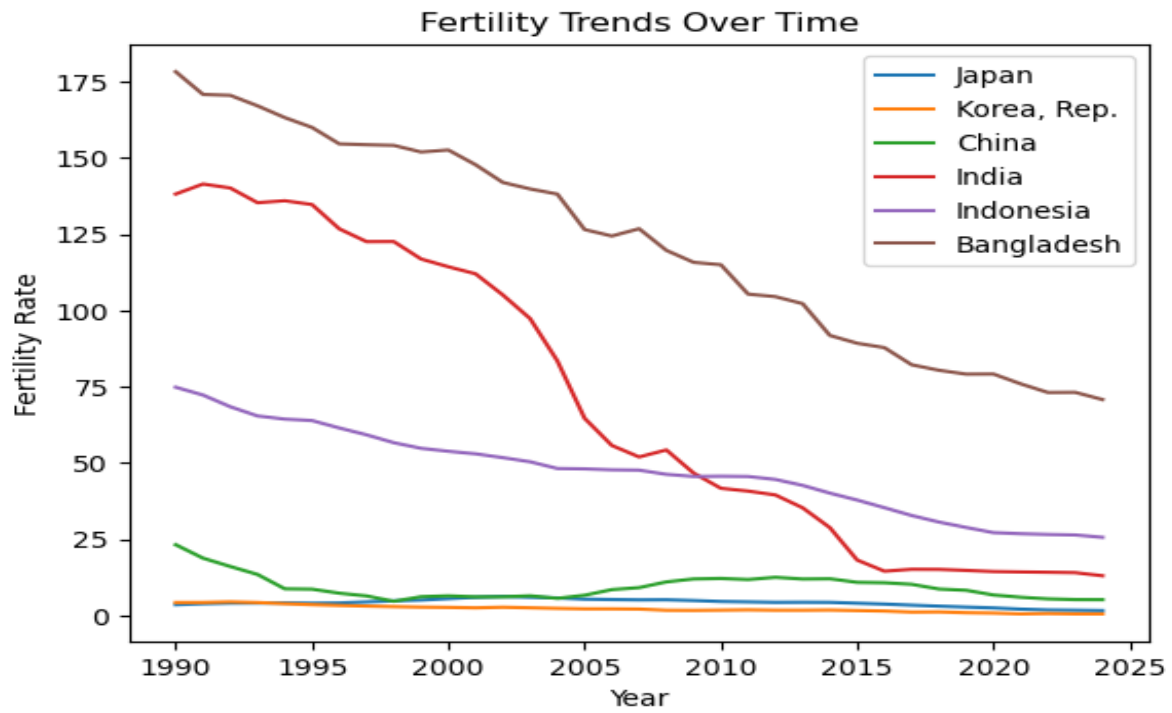


Figure 2. Adolescent fertility rate (births per 1,000 women aged 15–19), 1990–2024. Source: World Bank WDI (SP.ADO.TFRT).

3.3 Under-5 Mortality Rate Trends

Figure 3 illustrates the decline in under-5 mortality. The scale of improvement across South and Southeast Asia is extraordinary. Bangladesh fell from 146.5 deaths per 1,000 live births in 1990 to 30.6 in 2023 — a 79% reduction. India declined from 127 to 28, Indonesia from 83 to 18, and China from 53.6 to just 6.1. Japan and South Korea, already at very low levels in 1990, continued a gradual decline. The convergence in child mortality is even more pronounced than in life expectancy, indicating that targeted public health interventions — immunisation campaigns, oral rehydration therapy, and improved nutrition — have been remarkably effective across the region.

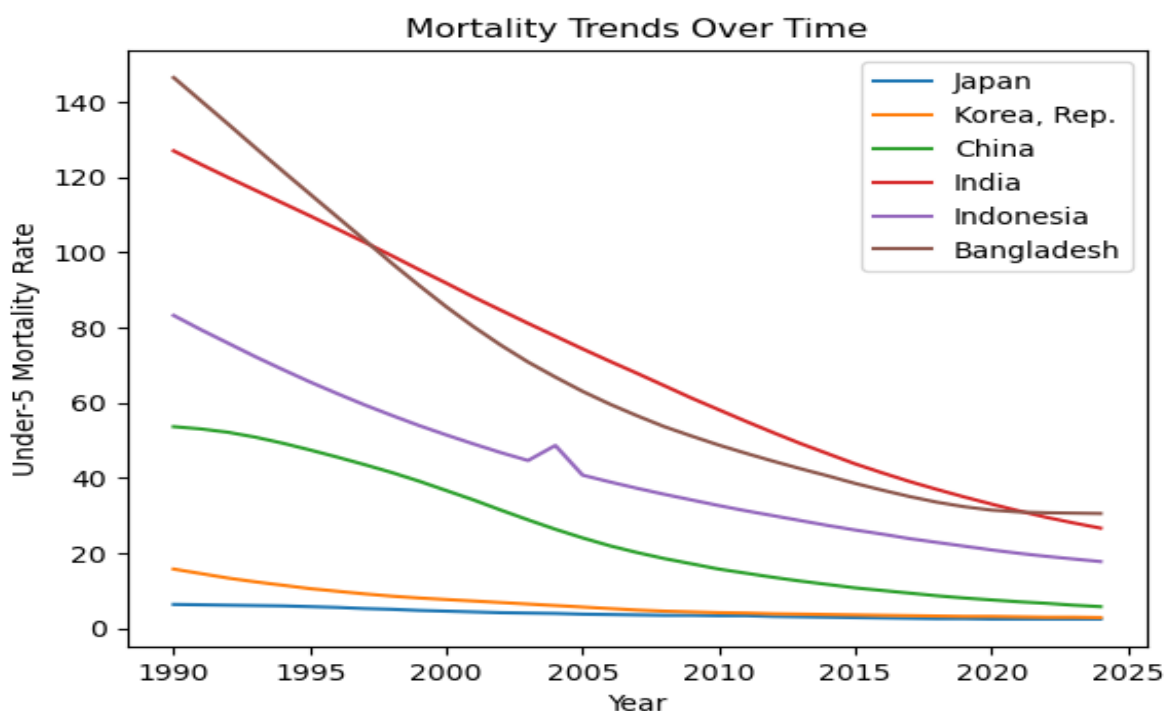


Figure 3. Under-5 mortality rate (per 1,000 live births), 1990–2024. Source: World Bank WDI (SH.DYN.MORT).

3.4 Relationships Between Indicators

Figures 4 and 5 explore the bivariate relationships between the three indicators, with each point representing one country-year observation.

Adolescent fertility and life expectancy (Figure 4). A strong negative association is apparent — country-years with higher adolescent fertility consistently show lower life expectancy. The temporal trajectory for each country runs from the upper-left corner (high fertility, low life expectancy in 1990) toward the lower-right (lower fertility, higher life expectancy by 2024). This is consistent with the demographic transition framework, in which lower fertility and better health reinforce each other through improved maternal and child outcomes, greater investment per child, and higher female labour force participation.

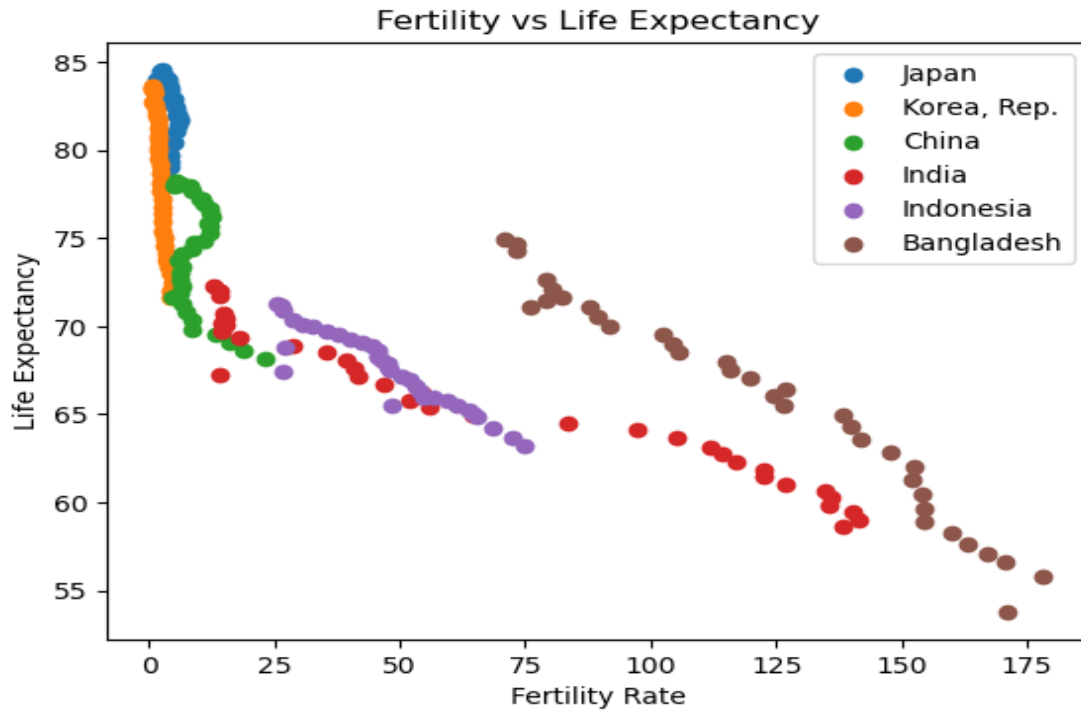


Figure 4. Adolescent fertility rate vs. life expectancy, 1990–2024. Each point represents one country-year observation.

Under-5 mortality and life expectancy (Figure 5). The negative relationship here is even tighter. Bangladesh and India — with mortality rates above 125 in 1990 — cluster at the low end of the life expectancy scale, while Japan and South Korea occupy the opposite corner. As child mortality declined over time within each country, life expectancy rose correspondingly, tracing diagonal trajectories toward the lower-right. This pattern underscores that child survival improvements are the single strongest driver of aggregate life expectancy gains in the region.

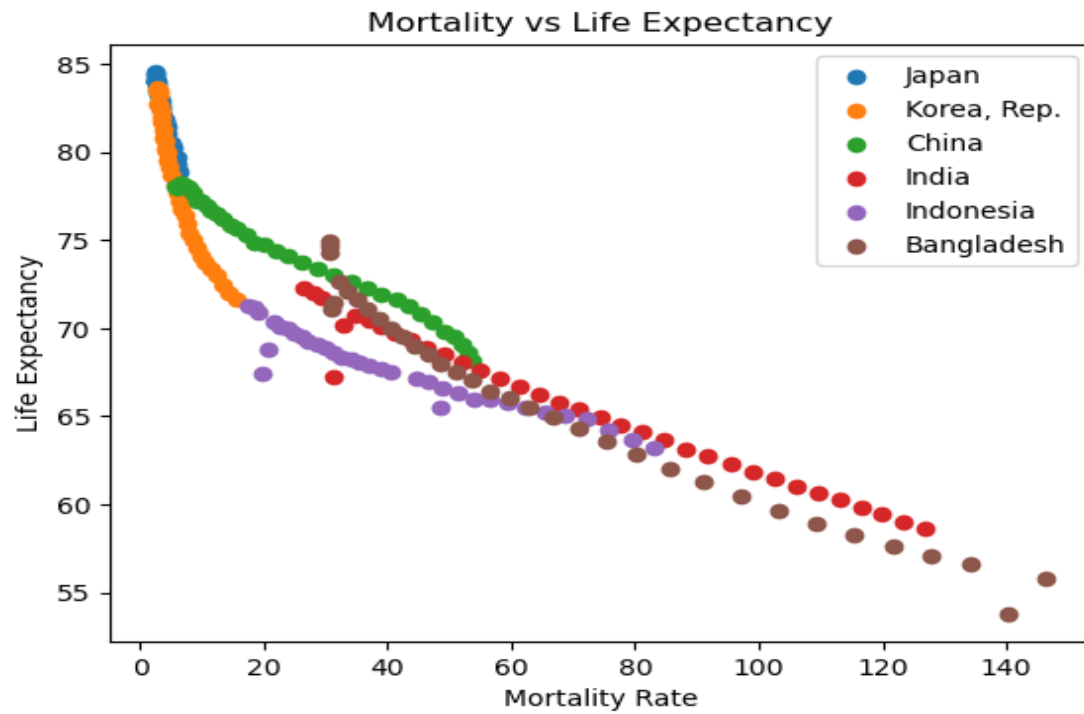


Figure 5. Under-5 mortality rate vs. life expectancy, 1990–2024. Each point represents one country-year observation.

4. Results and Discussion

Universal improvement across the region. Every country in the dataset registered gains in all three indicators between 1990 and 2024. This breadth is notable: it includes Pacific island micro-states, post-conflict economies such as Timor-Leste, and both high-income and lower-income nations. The consistency of improvement suggests that structural forces — economic growth, globalisation of medical knowledge, and international development aid — have operated broadly across the region.

South Asia's outsized progress. Bangladesh achieved the largest absolute life expectancy gain among the six focus countries (+18.9 years), driven by extraordinary reductions in child mortality (from 146.5 to 30.6 per 1,000) and adolescent fertility (from 178 to 73 per 1,000). This reflects decades of targeted investments by the government and NGOs — particularly in immunisation, family planning, and girls' education — and is well-documented as one of the most rapid public health advances in the development literature.

East Asian convergence. South Korea's life expectancy trajectory is the most remarkable in the region. Starting 7 years behind Japan in 1990, it has essentially closed the gap by 2023. This convergence is consistent with South Korea's sustained economic growth, universal health coverage, and very low fertility rates. The scatter plots confirm that South Korea has moved from a middle position to clustering with Japan at the healthiest end of the distribution.

Strong inter-indicator relationships. Both scatter plots confirm that child mortality and adolescent fertility are inversely related to life expectancy — and that the under-5 mortality relationship is tighter. This suggests that improvements in child survival are the primary lever for raising aggregate life expectancy, while adolescent fertility declines contribute through improved maternal health and better resource allocation per child.

Limitations

Data completeness varies across countries — particularly for small Pacific island states — and linear interpolation introduces smoothness that may mask real-world fluctuations. The analysis is descriptive and does not establish causality between indicators. Additionally, aggregating across countries with vastly different population sizes (China, 1.4 billion vs. Tuvalu, ~11,000) can be misleading when drawing regional generalisations. Future work could use population-weighted averages or panel regression methods to isolate the causal contributions of each indicator on the others.

5. Conclusion

This project analysed population dynamics across 36 Asian and Pacific countries using World Bank WDI data from 1990 to 2024. Combining SQL-based data processing with Python visualisation, we documented that life expectancy rose substantially, under-5 mortality fell dramatically, and adolescent fertility declined across the region — all consistent with an ongoing demographic transition.

The most striking finding is the speed and breadth of South Asia's improvement, particularly Bangladesh, where gains of 19 years in life expectancy and near-80% reductions in child mortality over just 33 years represent one of the most rapid public health advances on record. Meanwhile, East Asian high-income economies continue to push the frontier of human longevity, with South Korea nearly converging with Japan.

The strong negative relationships between child mortality and life expectancy, and between adolescent fertility and life expectancy, highlight the interconnected nature of demographic change. Policy efforts targeting child health and girls' education not only address those outcomes directly, but have multiplicative effects on overall population health. As the region continues to develop, these dynamics will shape labour markets, ageing populations, and social spending in the decades ahead.

Appendix: Reproducibility & Technical Notes

This report is fully reproducible. The complete workflow is contained in two Jupyter notebooks and one CSV data file, all included in the project repository.

File	Description
Final_project_group_4_1_.ipynb	Data acquisition from WDI API, cleaning, merging, and export to CSV
Final_project_group_4_with_SQL.ipynb	SQL descriptive statistics (SQLite) and all visualisations (Figures 1–5)
asian_population_clean.csv	Cleaned, merged dataset — 1,260 rows x 6 columns

Key SQL Queries

The following queries were run on an in-memory SQLite database populated from the cleaned CSV to produce the summary tables in this report.

```
-- 1990 vs 2024 Comparison
SELECT country,
  ROUND(MAX(CASE WHEN year=1990 THEN life_expectancy END),2) AS life_1990,
  ROUND(MAX(CASE WHEN year=2024 THEN life_expectancy END),2) AS life_2024,
  ROUND(MAX(CASE WHEN year=2024 THEN life_expectancy END)
    - MAX(CASE WHEN year=1990 THEN life_expectancy END),2) AS life_gain
FROM asian_data GROUP BY country ORDER BY life_gain DESC;

-- Overall statistics per country
SELECT country,
  COUNT(DISTINCT year) AS years_covered,
  ROUND(AVG(life_expectancy),2) AS avg_life_expectancy,
  ROUND(MAX(life_expectancy)-MIN(life_expectancy),2) AS life_improvement,
  ROUND(AVG(mortality_u5),2) AS avg_mortality_u5,
  ROUND(AVG(fertility_adolescent),2) AS avg_teen_fertility
FROM asian_data GROUP BY country ORDER BY avg_life_expectancy DESC;
```

Python Libraries

Library	Purpose
pandas	Data manipulation, merging, and interpolation
requests	World Bank WDI API access
sqlite3	In-memory SQL queries (standard library)
matplotlib	All five visualisations (Figures 1–5)